

COLUMN: AI EXPERT

The DAO to DeSci: AI for Free, Fair, and Responsibility Sensitive Sciences

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This article discusses the impact and significance of the autonomous science movement and the role and potential uses of intelligent technology in DAO-based decentralized science (DeSci) organizations and operations. What is DeSci? How does it relate the science of team science? What are its potential contributions to multidisciplinary, interdisciplinary, and/or transdisciplinary studies? Does it have any correspondence to the social movement organizations in traditional social sciences or the cyber movement organizations in the new digital age? Particularly, issues on DeSci to current professional communities, such as IEEE and its societies, conferences, and publications, are addressed, and the effort for the framework and process of DAO-based DeSci for free, fair, and responsibility sensitive sciences is reviewed.

The autonomous science, particularly decentralized autonomous organizations (DAO)-based decentralized science (DeSci), is emerging as an important movement in research and development in recent years.¹⁻³ To investigate its potential significance, on March 21 and 30, 2022, our AI Expert of the *IEEE Intelligent Systems* has held two decentralized hybrid workshops (DHWs) to address various issues of DeSci and its impact on the existing ecology and future evolution of scientific activities and operations. Specifically, what is DeSci? What is its relationship to the science of team science (SciTS), social movement organizations, cyber movement organizations, and multi/inter/transdisciplinary studies? Impact and significance by examples?

Immediate and potential impacts to organizations and operations of traditional professional communities, such as IEEE? What is the role of AI in DeSci? And finally, what should we do next? This article summarizes briefly our discussion in the DHWs and the correspondence among some of the participants of the Free, Fair, and Responsibility Sensitive Sciences (F2rS2) project. A decentralized hybrid symposium on DeSci will be scheduled late this year and an in-depth report will be published accordingly.

DAO-BASED DESCI: WHAT IS AND WHAT IN IT?

It is hard to specify precisely a rapidly evolving field and its corresponding movement, but DeSci's two defining features are self-evident: 1) crowdsourcing and cyber-based decentralized organizations for its activities, and 2) blockchain and web3 (or Web 3.0) supported operations for its management.¹⁻⁴ Its obvious motivation and objective are the creation of new mechanisms and processes for raising and distributing funds for

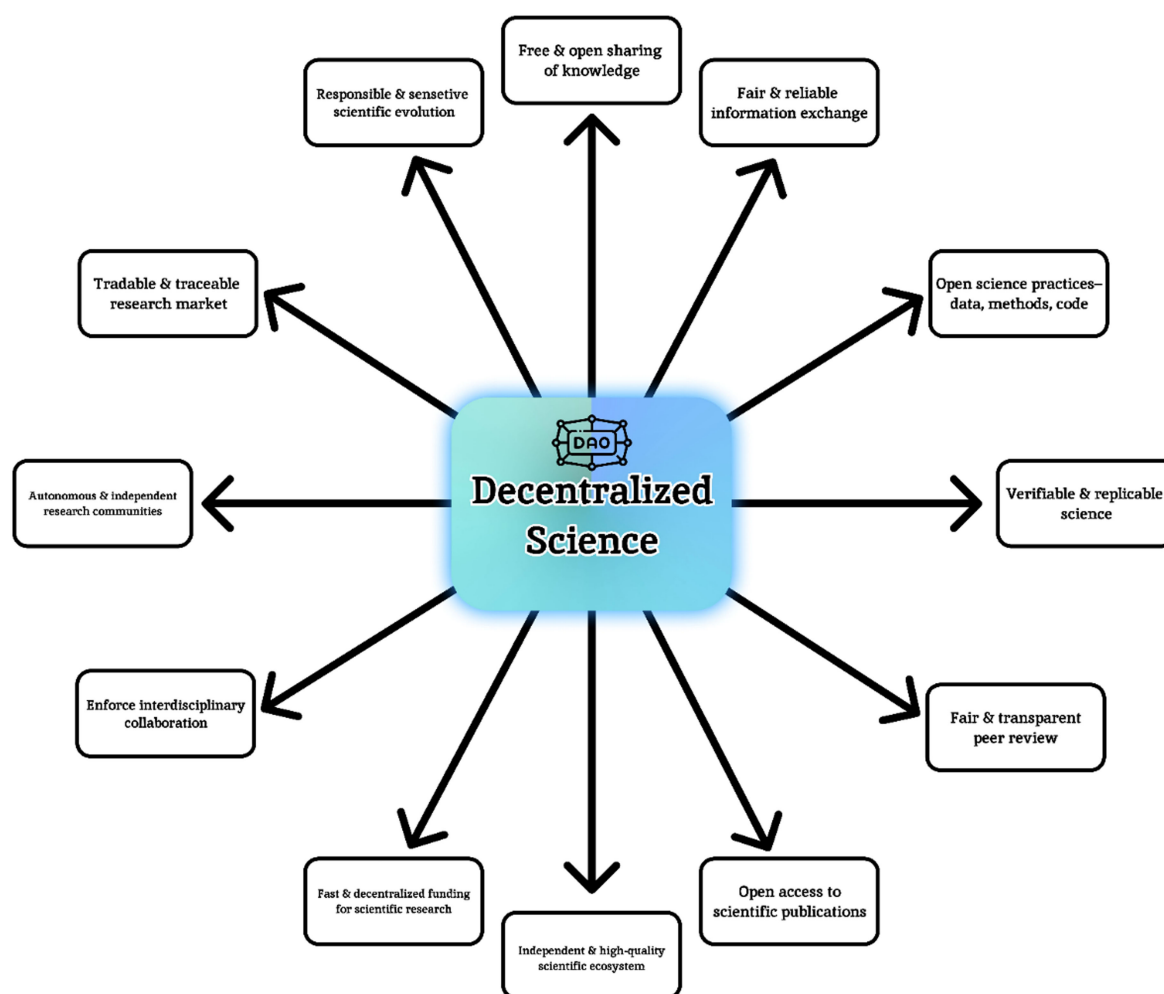


FIGURE 1. Basic characteristics and indicators of DeSci.

scientific endeavors and generating and disseminating knowledge for common goods, particularly, shifting “ownership and value away from industrial intermediaries” and removing “research pain points, silos, and bottlenecks.”^{2,3} However, this movement is fundamentally a natural consequence of the need for multi/inter/transdisciplinary studies, the emergence of past social movement organizations and recent cyber movement organizations, the call for the SciTS, and the convergence of open science and AI-oriented open-source movements.^{1,5} Its recent fast development and deployment are mainly driven by the success and publicity of web3, crypto technology, and blockchain intelligence, especially smart contracts, and decentralized autonomous organizations and operations.⁴

Figure 1 illustrates the essential characteristics and indicators of DeSci. To a large extent, the current DeSci is just a set of mechanisms and supporting

infrastructures for bottom-up individual sensemaking through the use of blockchain and web3 technologies. Its main driving force is generated by web3 collectives and blockchain-based DAOs. Specifically, DeSci can be considered as the use of smart contracts and tokens (fungible and nonfungible) to open up markets through the deployment of open-source finance tools, especially decentralized finances (DeFi).

DeSci is part of a bigger movement using recent digital tools for funding, organizing, training, planning, coordinating, dispatching, collecting, distributing of supply-demand activities, and resources in cyberspace-based communities.^{6–10} From a narrow perspective, DeSci is the capacity of individual agents or specialized communities to make sense of the world autonomously, by defining their own problems, tasks, languages, imaginations, philosophies, and methodologies. From a broader perspective, DeSci is the emergence of collective desires

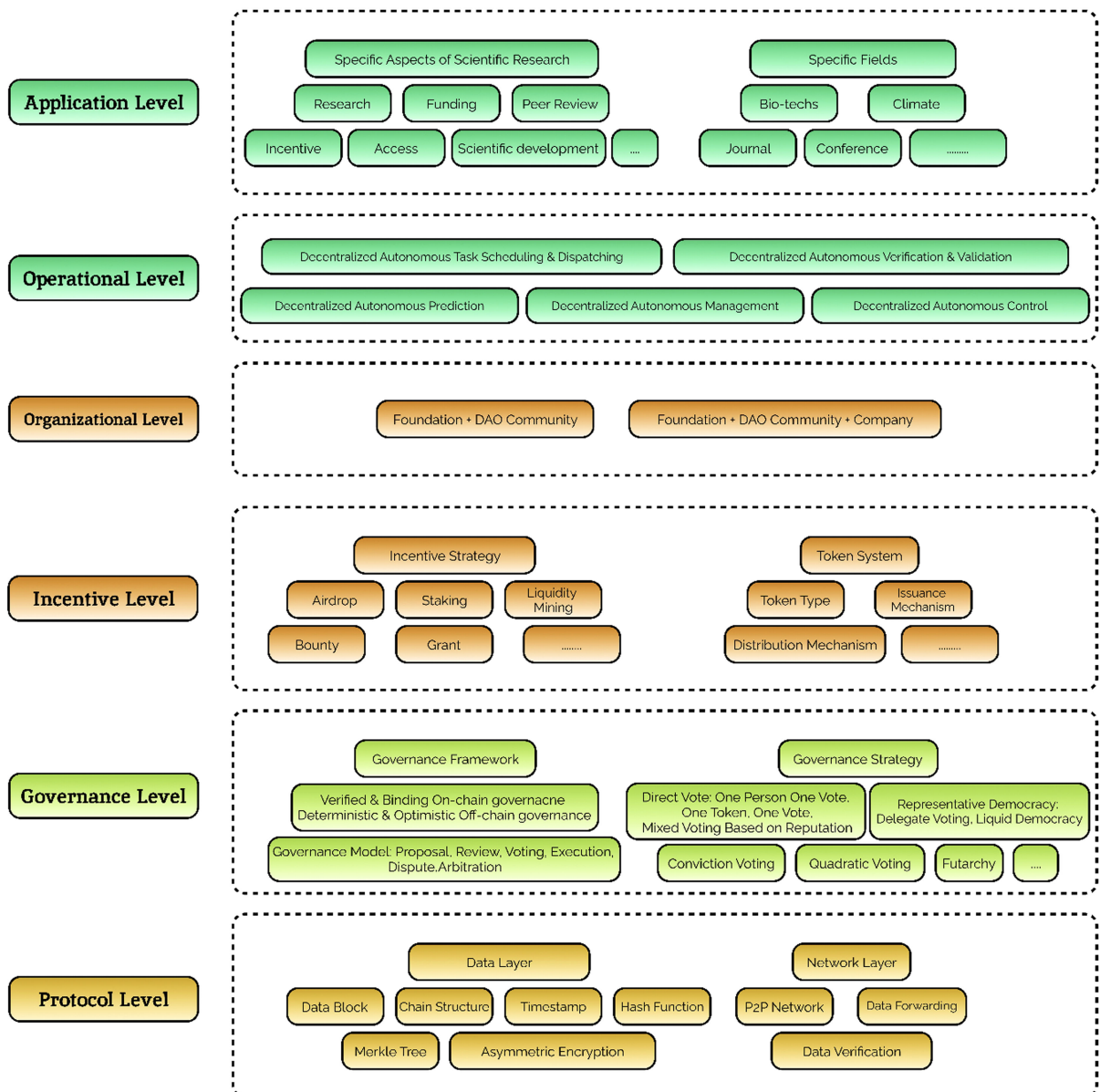


FIGURE 2. Reference model for DeSci.

to improve the research and development establishments, which are expected to be transformed by intelligent technologies into more effective and efficient organizations with increased ability for sciences to fulfill their mandates and align with human values.¹⁻⁵

In traditional centralized science (CeSci), centralized agents, such as government institutions, philanthropic foundations, private businesses, or universities, have been organized and operated for pragmatic rather than purely epistemic purposes, where techno-scientific institutions have a core mandate of helping outside institutions in their pragmatic endeavors with an overarching goal of

accumulating epistemic value along this direction. Figure 2 presents the framework for a DeSci model derived from the initial idea in Wang's work.¹ Clearly, DeSci differs from CeSci in terms of its tools, structure, governance, incentives, organization, operations, and norms, or more generally, in terms of its research and development ecology for knowledge discovery and utilization.

MOLECULE AND ITS IPNFT: A CASE OF BIOTECH DAOS FOR DESCI

Molecule is a decentralized underlying infrastructure and marketplace for the biomedical field. Early stage

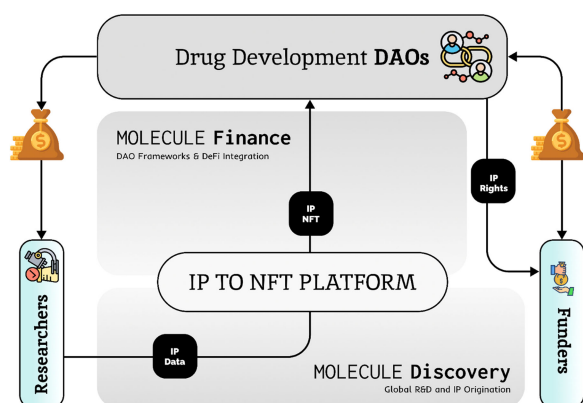


FIGURE 3. Operational process of Molecule.

biopharmaceutical research is the lifeblood of advancing medicine. However, many areas of biopharmaceuticals typically face three problems: chronic underfunding or challenges of commercialization, legal complexity, illiquidity of intellectual property assets, and lack of connection to relevant investors.

To solve the abovementioned problems, Molecule builds an open market and underlying infrastructure by applying web3-related technologies, making new research and IP discoverable, generally available, and assigning ownership to participants. As shown in Figure 3, the Molecule protocol consists of three parts in the following.⁶

IP-NFT: A new type of asset class, namely virtual asset IP. It virtualizes and modularizes IP by combining legal and technical frameworks with NFT technology. Due to the complexity of IP in the medical field, IP-NFTs need to satisfy these characteristics: privacy, permanence, accessibility, and verifiability. The implementation of the abovementioned features of IP-NFT is based on Ethereum to ensure the ownership of assets, the metadata storage layer is based on Arweave to ensure that data are permanently available, and the data storage layer is based on the Nevermined platform to ensure the accessibility and verifiability of data. In addition, to achieve the tradability of IP-NFT and accelerate its liquidity, unique digital asset identifiers—NFT, new automatic exchange infrastructure—automatic market maker, and governance structure—DAOs to reshape intellectual property ownership, financing, and entity creation.

Molecule discovery: It is a platform for intellectual property and data interaction. Researchers can make their biopharmaceutical assets (data and preapplied patents) visible to everyone, receive feedback, and interact with potential funders. When potential investors decide to invest in the IP, the IP will be transferred

in the form of NFT. The mission of molecule discovery is to accelerate the discovery of promising early-stage treatments and bridge the “valley of death” in academia.

Molecule finance: IP-NFTs need a high liquidity pool to function. Molecule finance enables modular drug development by creating a virtual funding and collaborative environment. NFTs can be transferred to automated market makers to raise capital or inserted into DAOs managed as portfolios for specific therapeutic verticals. This model reflects a virtual biopharmaceutical startup with global stakeholders, funders, and researchers, but assigns governance rights entirely to the community.

Molecule enables biopharmaceutical research and related intellectual property to be funded as NFT for the first time. It provides a new reference model for development in the field of biotechnology. Currently, the Molecule platform has funded several projects. For example, *VitaDAO* focuses on longevity research, *PsyDAO* on psychedelic drug research, and the biotech community *LabDAO*.

In addition to Molecule, the application scenarios of DeSci are also expanding in two specific directions. General-purpose issues, such as decentralized funding,⁷ peer review,⁸ access, and scientific development,⁹ and research and development in specific fields, such as environmental studies.¹⁰ Clearly, the DeSci movement is moving from theoretical ideas and small-scale technical experiments to more established players for funding university research and launching multiple DAOs.

IMPACT AND SIGNIFICANCE: SOCIETIES, CONFERENCES, AND PUBLICATIONS

The case of Molecule as a Biotech DAO in *OpenSea* has demonstrated the initial vitality and success of the DeSci movement.⁶ The original goal and function of its IPNFT are to use IP legal languages and procedures for helping NFT creators and collectors protect their rights more effectively and efficiently, but vice versa are also true, i.e., NFT and other tokens can be used to help IP developers and owners better secure and monetize their rights. This would enable a new kind of ownership, particularly make communities be the new shareholders of scientific knowledge, and eventually build self-sustaining scientific ecosystems where values generated by knowledge assets can be used to fund the creation and utilization of new knowledge.³

The impact of experimenting new cyber tools and AI technologies to improve scientific knowledge generation

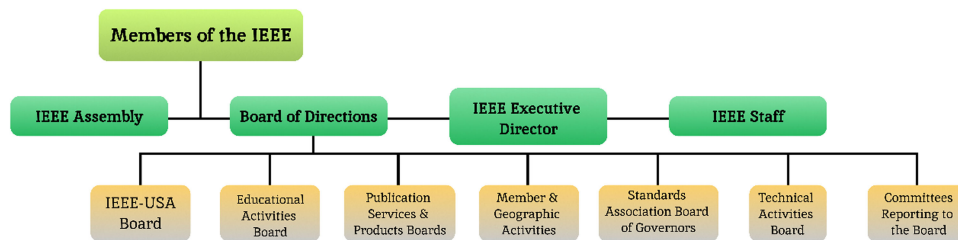


FIGURE 4. IEEE structure and activities.

and distribution, and ultimately move toward knowledge automation, such as the effort by DeSci, will be significant but uncertain in its scale and sustainability at this early stage. A major issue here is that the DeSci movement is happening with none or little input from established scientific communities so far. We have to ask ourselves: Is DeSci a disruptive technology for knowledge industries? Is DeSci movement a revolution for scientific establishments? Specifically, what is its impact on IEEE? Still the world's largest technical professional organization for advancing technology for humanity in a possible age of coming to DeSci?

Figure 4 outlines the IEEE organizational structure and activities. With a long history starting from the American Institute of Electrical Engineers in 1884 and over 400,000 members in 160 countries and regions, IEEE is well known for its professional societies/councils, technical conferences, academic publications, and industrial standards. According to IEEE 2020 Annual Report, it has over five million documents with total usage of 192,262,982, including 222,035 new articles from 1611 IEEE sponsored conferences with over 465,000 attendees in 96 countries, 86,052 new journal and magazine papers, and 138 standards approved for publication, and stands as a financial giant among professional communities worldwide with a total asset over \$917 million, revenue over \$467 million, and expense over \$397 million in 2020.

In the past decade, in addition to other professional organizations, IEEE is facing serious competition from commercial publishers and open-source movements for social influences, asset accumulations, and financial benefits. The strength and foundation for IEEE growth lie significantly in its services and contributions from member volunteers; however, this might also turn to be IEEE's Achilles' facing the autonomous science movement, since DeSci offers an attractive alternative and a potential paradigm shift for volunteer services and contributions.

Therefore, IEEE must investigate, test, and evaluate the potential of DeSci for its organizational and operational procedures and activities as soon as

possible. For example, DeSci will help IEEE in its collection, distribution, and even monetization of articles for both conferences and publications, its event organization and coordination, and its financial inspection and accountability, as well as its creation of a new and verifiable reputation and honor systems for its members. A list of topics will be addressed and discussed in the coming IEEE DHS-DeSci.

F2RS2: A PROJECT FOR FREE, FAIR, AND RESPONSIBILITY SENSITIVE SCIENCES

Figure 5 illustrates the primitive structure of a proposed project for "Free, Fair, And Responsibility Sensitive Sciences (F2rS2)" with a new kind of ownership that would be neither public owed nor private owed, called Responsibility Sensitive Ownership (RSO) using blockchain intelligence and technology.¹ The F2rS2 was initially launched in 2016 by the Qingdao Academy of Intelligent Industries with the support of the UNDP

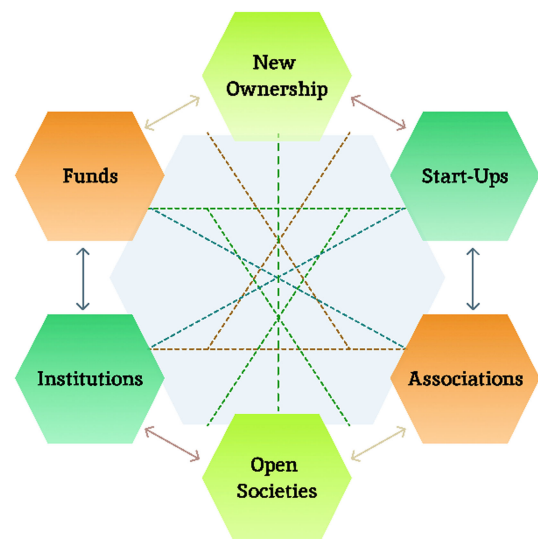


FIGURE 5. F2rS2 elements and structure: Open societies and RSO.

in Thailand and the help of the Chinese Association of Automation (CAA), International Federation of Automatic Control (IFAC) Technical Committee on Social and Economic Systems, as well as the Beijing Chapters of International Council on Systems Engineering, Association for Computing Machinery, and Association for Advancement of Artificial Intelligence. The motivation was to establish a new research and development ecosystem with new digital and smart tools in cyber-physical-social spaces (CPSS), an end-to-end whole chain, and seamless organizational and operational support for professional communities from open societies, academic associations, research institutes, startups, capital funds, to new ownership, RSO.

As a pilot project, the Association of Intelligent Science and Technology and the Association for Advancement of Intelligent Industries were registered at Denver, CO, USA, in 2016 under the leadership of Fei-Yue Wang and Nils Nilsson, which were merged into a single entity, the Association for Global Intelligent Science and Technology (AGIST) after the death of Nilsson. A workshop called Blockchain and Knowledge Automation has been launched in 2017 at Denver University and became a conference in 2018 under the support and sponsorship of IEEE, IFAC, and CAA. In 2019, *Chinese Journal of Intelligent Science and Technology* was launched, after the failure of establishing it as an IEEE/CAA joint publication, and in 2021, CAA restarted *International Journal for Intelligent Control and Systems* (established in 1996) and AGIST launched International Conference of Cyber-physical and Social Intelligence with IEEE. Those associations, publications, and conferences have been designed and would be deployed for DAO-based "mosaic organizations" with "mosaic operations" for "mosaic events" in building the F2rS2 project.

The goal of F2rS2 is to establish an open-source platform to support professional communities with a new kind of ownership for responsibility sensitive scientific activities of association, publication, and conference using new digital tools and smart methods in AI and intelligent technology, in an attempt to facilitate the development of self-sustaining scientific ecosystems. Since 2016, it has been supported entirely by individual volunteers and NGOs and has attracted the funds from capitals for AI as well as projects from traditional industries, such as mining, manufacturing, semiconductors, power and energy, and robotics and automation.

CALL FOR PARTICIPATION AND DISCUSSION: DHS ON DESCI

IEEE IS AI Expert will organize a *Decentralized Hybrid Symposium on DeSci* (DHS-DeSci) starting from June

to August this year. *DHS-DeSci* will focus on the following:

- 1) The State and Trend of DeSci.
- 2) Open Science, SciTS, CMO, DAO, Metaverse, Web 3.0, CPSS.
- 3) DeSci and DAO for Professional Communities and Associations.
- 4) DeSci and DAO for Academic Journals and Publications.
- 5) DeSci and DAO for Conferences and Academic Activities.
- 6) New IT and Open Infrastructure for F2rS2 Project, etc.

A series of decentralized hybrid workshops on different topics will be held during *DHS-DeSci* in Asia, Europe, and North America, respectively. Our discussions will be summarized and reported at AI Expert. A position paper on DeSci is expected in the end of *DHS-DeSci*.

Looking forward to having you in *DHS-DeSci*.

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